

ON THE NON-CENTRAL DISTRIBUTION OF THE LARGEST CANONICAL CORRELATION COEFFICIENT

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1. Introduction

The distribution of the canonical correlation coefficients $r_1^2, r_2^2, \dots, r_q^2$ ($0 < r_q^2 < \dots < r_1^2 < 1$) between variates $y_1, \dots, y_q$ and $x_1, \dots, x_r$ ($q \leq r$) calculated from a sample of $N = n - 1$ observations depends on the population canonical correlation coefficients $\rho_1^2, \rho_2^2, \dots, \rho_q^2$ and is (Constantine (1963)),

$$(1.1) \quad C_1(q, r, n) |I - P|^{n/2} |R|^{(r-q-1)/2} |I - R|^{(n-r-q-1)/2} a_q(R) \cdot {}_2F_1(n/2, n/2; r/2; P, R).$$

$$P = \text{diag}(\rho_1^2, \dots, \rho_q^2), \quad R = \text{diag}(r_1^2, \dots, r_q^2), \quad a_q(R) = \prod_{i=1}^q (1 - r_i^2)^{-1/2},$$

and

$$C_1(q, r, n) = \pi^{q^2/2} \Gamma_q(n/2) / \Gamma_q(q/2) \Gamma_q(r/2) \Gamma_q((n-r)/2).$$

${}_2F_1$ is the hypergeometric function of matrix argument, (James (1964)), given by

$$\sum_{k=0}^{\infty} \frac{\Sigma_K(n/2) \Sigma_K(n/2) C_K(P) C_K(R)}{\Sigma_K(r/2) \Sigma_K((n-r)/2)} C_K(I_q).$$

(K being a partition of k .)

The purpose of this note is to derive the distribution of r_1^2 and to give a similar result for the distribution of the largest root of the matrix in multivariate analysis.

2. The distribution of the largest canonical correlation coefficient

Let $x_{i-1} = r_i^2 / r_1^2$ ($i = 2, \dots, q$), $X = \text{diag}(x_1, x_2, \dots, x_{q-1})$ and $X^1 = \text{diag}(1, x_1, \dots, x_{q-1})$ in (1.1) after using the expression, (Khatri and Pillai (1968)),

$$|I - R|^{(n-r-q-1)/2} C_K(R) = \sum_{j=0}^{\delta} \sum_J \sum_{\delta} g_{K,J}^{\delta} ((r-n+q+1)/2)_{\delta} C_{\delta}(R) / j!$$

(J being a partition of j .)

$g_{K,J}^{\delta}$ is a constant and has been tabulated for various values of the three arguments. δ is a partition of $k+j$ into not more than q components.

The joint distribution of r_1^2 and X can therefore be written as

$$(2.1) \quad C_1(q, r, n) |I - P|^{n/2} |X|^{(r-q-1)/2} |I - X| a_{q-1}(X) \sum_k \sum_K \sum_j \sum_J g_{K,J}^{\delta} \\ r_1^{2(rq/2+k+j-1)} (n/2)_K (n/2)_K ((r-n+q+1)/2)_J C_K(P) C_{\delta}(X^1) \\ [(r/2)_K C_K(I_q) k! j!]^{-1}$$

Integrating with respect to X using the integral (Sugiyama (1967)),

$$\int_0 < x_1 < \dots < x_{q-1} < 1 |X|^{(r-q-1)/2} |I - X| a_{q-1}(X) C_{\delta}(X^1) dX$$

$$= \Gamma_q(q/2) \Gamma_q(r/2) \Gamma_q((q+1)/2) (rq/2+k+j)(r/2)_\delta C_\delta(I_q) / (\pi^{q^2/2} \Gamma_q((r+q+1)/2) ((r+q+1)/2)_\delta)$$

we obtain the distribution of r_1^2 , i.e.

$$(2.2) \quad C_2(q, r, n) |I-P|^{n/2} \sum_k \sum_K \sum_j \sum_J \sum_\delta r_1^2 (rq/2+k+j-1) (rq/2+k+j) A_{K,J}^\delta(P)$$

where

$$(2.3) \quad C_2(q, r, n) = \Gamma_q(n/2) \Gamma_q((q+1)/2) / \Gamma_q((n-r)/2) \Gamma_q((r+q+1)/2)$$

and

$$(2.4) \quad A_{K,J}^\delta(P) = g_{K,J}^\delta(n/2) K(n/2) K(r/2)_\delta ((n-r+q+1)/2) J C_K(P) C_\delta(I_q) / (r/2)_K ((r+q+1)/2)_\delta C_K(I_q) k! j!.$$

The cdf of the largest canonical correlation coefficient r_1^2 is therefore given by

$$(2.5) \quad P(r_1^2 \leq z) = C_2(q, r, n) |I-P|^{n/2} \sum_k \sum_K \sum_j \sum_J \sum_\delta z^{rq/2+k+j} A_{K,J}^\delta(P).$$

If P is of rank one, i.e. ρ_1^2 is the only population canonical correlation coefficient, then (James (1961)),

$$C_K(\rho_1^2) = (\rho_1^2)^k.$$

Hence, 2.5 becomes

$$(2.6) \quad P(r_1^2 \leq z) = C_2(q, r, n) (1-\rho_1^2)^{n/2} \sum_k \sum_K \sum_j \sum_J \sum_\delta z^{rq/2+k+j} A_{(k),J}^\delta(\rho_1^2),$$

where $K = (k)$ is now the only partition k .

3. The distribution of the largest root of a matrix

Consider the following joint distribution of the roots $r_1, r_2, \dots, r_p$ ($0 < r_p < \dots < r_1 < 1$) of a matrix in multivariate analysis, (Constantine (1963),

$$(3.1) \quad C_3(p, m, n) \exp(-\text{tr } \theta) |\Delta|^{(m-p-1)/2} |I-\Delta|^{(n-p-1)/2} a_p(\Delta)$$

$${}_1F_1((m+n)/2; m/2; \theta, \Delta)$$

where $\Delta = \text{diag}(r_1, \dots, r_p)$

$$\text{and } C_3(p, m, n) = \pi^{p^2/2} \Gamma_p((m+n)/2) / \Gamma_p(p/2) \Gamma_p(m/2) \Gamma_p(n/2).$$

θ is the non-centrality parameter.

Khatri and Pillai (1968) derived the distribution of r_1 in this case. Following the method given in (2) however, the distribution of r_1 can be written in a form more convenient for calculation purposes, namely

$$(3.2) \quad C_4(p, m, n) \exp(-\text{tr } \theta) \sum_k \sum_K \sum_j \sum_J \sum_\delta r_1^{mp/2+k+j} (mp/2+k+j) B_{K,J}^\delta(\theta)$$

where

$$C_4(p, m, n) = \Gamma_p((m+n)/2) \Gamma_p((p+1)/2) / \Gamma_p(n/2) \Gamma_p((m+p+1)/2)$$

and

$$B_{K,J}(\theta) = g_{K,J}^{\delta}((m+n)/2, K((n+p+1)/2), J(m/2))_{\delta} C_K(\theta) C_{\delta}(I_p) / \\ (m/2)_{K((m+p+1)/2)}_{\delta} C_K(I_p) k! j!$$

The cdf of r_1 can therefore be written as

$$(3.3) \quad P(r_1 \leq z) = C_4(p, m, n) \exp(-\text{tr} \theta) \Sigma_K \Sigma_K \Sigma_J \Sigma_J \Sigma_{\delta} z^{mp/2+k+j} B_{K,J}^{\delta}(\theta).$$

If θ is of rank one, then there exists an orthogonal matrix H such that $H'\theta H = \text{diag}(\lambda, 0, \dots, 0)$ and hence (3.3) becomes

$$(3.4) \quad P(r_1 \leq z) = C_4(p, m, n) \exp(-\lambda) \Sigma_K \Sigma_J \Sigma_J \Sigma_{\delta} z^{mp/2+k+j} B_{(k),J}^{\delta}(\lambda).$$

REFERENCES

- CONSTANTINE, A.G. (1963). Some non-central distribution problems in multivariate analysis. *Ann. Math. Statist.* **34**, 1270-1285.
- JAMES, A.T. (1961). The distribution of non-central means with known covariance. *Ann. Math. Statist.* **32**, 874-882.
- JAMES, A.T. (1964). Distributions of matrix variates and latent roots derived from normal samples. *Ann. Math. Statist.* **35**, 475-501.
- KHATRI, C.G. and PILLAI, K.C.S. (1968). On the non-central distribution of two test criteria in multivariate analysis of variance. *Ann. Math. Statist.* **39**, 215-226.
- SUGIYAMA, T. (1967). Distribution of the largest root of the generalised B statistic and F statistic in multivariate analysis. *Ann. Math. Statist.* **38**, 1152-1159.

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